Fixing the Dark-Energy Equation of State with Two Dimensionless Constants, φ and π: A Falsifiable, Parameter-Frugal Test Against DESI DR2 and Planck PR4

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Abstract

We present a minimal-parameter framework for the dark sector in which the *shape* of late-time cosmic acceleration is fixed algebraically by two dimensionless constants, the golden ratio $\varphi = (1+\sqrt{5})/2$ and π, rather than by tuned cosmological parameters. The equation of state follows as bare ratios of structural invariants, $w_0 = -T_r/M_v = -0.840$ and $w_a = -P_{\rm sc}/I_g = -0.670$, with no fitted amplitude. The point was fixed in advance and survives *two independent* DESI releases: it sits at 0.05σ from the DESI DR1 w_0w_aCDM posterior and, after the uncertainties tightened, at 0.24σ from the newer DESI DR2 posterior (Pantheon+; 0.2–1.8σ across supernova compilations, $\chi^2_{\rm 2D} = 0.42$)—the prediction did not move while the data improved. Crucially, the two forward predictions that can *refute* the core framework—this fixed (w_0, w_a) point and the tensor ratio $r = \varphi^{-10}$—stand *independent* of the φ-dark-matter extension: neither the Hubble resolution nor the growth sector is invoked to state or to kill them. A Markov-chain analysis combining DESI BAO with the ω_m-direct CMB anchor as prior yields $H_0 = 67.95 \pm 0.40$ km s^{-1} Mpc^{-1} (0.88σ from Planck, 0.04σ from the algebraic anchor). The effective-field-theory description is canonical—$\alpha_T = \alpha_M = \alpha_B = 0$ with $\alpha_K(0) = 0.4033$—and the inflationary sector predicts $n_s = 1 - \varphi^{-7} = 0.9656$ and a tensor-to-scalar ratio $r = \varphi^{-10} = 8.1 \times 10^{-3}$ that is testable by LiteBIRD and CMB-S4. A Boltzmann calculation shows the full algebraic ω_m (not the bare dynamical $\Omega_{m,\rm dyn} = 0.160$) is physically required for the CMB: without it the acoustic-peak structure degrades by a factor of ~ 22 in RMS. We are explicit about the framework's limits: the absolute vacuum scale enters as a single measured input, the single-sector growth amplitude S_8 remains in tension with weak-lensing surveys (the two-sector extension that resolves it is kept outside the core register), and the algebraic origin of the CMB matter density (ω_b, ω_c) together with the φ-dark-matter mass coefficient are open problems. Both core forward predictions bear on *unreleased* data: the (w_0, w_a) point is pre-registered to be excluded at $> 3\sigma$ if DESI DR3 shifts it, and $r = \varphi^{-10} = 8.1 \times 10^{-3}$ is testable by LiteBIRD—either alone can refute the framework without touching the extension sector. The result is a falsifiable, parameter-frugal description of dark energy whose core predictions rest on φ and π alone.

Contents

Notation: algebraic constants used in this work

All constants below are *dimensionless* and built solely from the golden ratio $\varphi = (1+\sqrt{5})/2$ and π. Constants defined in earlier SSEE material but *not used* in this submission (e.g. VITA) are deliberately omitted to keep the register honest.

Symbol	Definition	Value	Role
φ	$(1+\sqrt{5})/2$	1.61803	Golden ratio (UV seed)
π	—	3.14159	Circle constant (IR seed)
Ω	$\varphi+\pi$	4.75963	Stability metric
β (BIAL)	$(\varphi+\pi)/2$	2.37981	Base coupling scalar
KAL	$\beta+\pi$	5.52141	Structural viscosity
AURA	$\beta+\varphi$	3.99785	Screening amplitude
MIRA	AURA/2	1.99892	Mirror Ratio (enters f_{screen}, Paper 9)
P_{sc}	$\Omega+\varphi$	6.37766	Dynamical evolution scalar
I_g (IGNIS)	$\pi+P_{\text{sc}}$	9.51925	Disruptive convergence (w_a denominator)
K_v	$\varphi+\pi+\Omega$	9.51925	Structural convergence (builds M_v; distinct lineage from IGNIS)
T_r	$3(\varphi+\beta)$	11.99354	3D saturation horizon
M_v	$\varphi+\pi+K_v$	14.27888	Maximal dimensional invariant

Derived background quantities (all dimensionless):

$$w_0 = -\frac{T_r}{M_v} = -0.83995, \qquad w_a = -\frac{P_{\text{sc}}}{I_g} = -0.66997, \qquad \Omega_{\text{m}}^{\text{dyn}} = 1 + w_0 = 0.16005.$$

1 Introduction

The standard cosmological model, ΛCDM, fits the available data with six free parameters [Planck Collaboration, 2020], but pays for this success with a dark sector whose composition and energy scale remain unexplained. The recent DESI DR2 measurements, which favour a dynamical, evolving dark-energy equation of state over a pure cosmological constant [Abdul-Karim et al., 2025], sharpen the question: if dark energy is dynamical, what fixes its evolution? A theory that could *predict* the equation of state, rather than fit it, would mark genuine progress.

This paper develops one such proposal. The central hypothesis of the Structural Self-Energy Expansion (SSEE) framework is that the *shape* of the dark sector—the equation-of-state ratios, the effective-field-theory function profile, and the two independent matter fractions (dynamical and cosmological)—is fixed algebraically by two dimensionless constants: the golden ratio $\varphi = (1+\sqrt{5})/2$ and π. From these, a small set of structural invariants (Table) is built, and the late-time observables follow as bare ratios with no tuned amplitude.

We are deliberate about what this does and does not claim. SSEE is a *minimal-parameter* framework, not a zero-parameter one: the absolute scale of the vacuum enters as a single measured input (Postulate D, §7), reducing the count of free CMB inputs from ΛCDM's six to two (A_s and τ; $k=2$ vs $k=6$). The dark-sector *shape*, however, carries no free parameter, and this is what makes the framework falsifiable. Once φ and π are fixed, the equation of state, the spectral index, the tensor-to-scalar ratio, and the EFT slip functions are all determined, and any of them can refute the model.

The paper is organised as a single coherent argument rather than a collection of results. Section 2 constructs the equation of state from the algebraic invariants and confronts it with DESI DR2, derives the two-Ω structure that distinguishes the dynamical matter fraction $\Omega_{\text{m}}^{\text{dyn}} = 0.160$ from the cosmological $\Omega_{\text{m}}^{\text{cosm}} = \omega_m/h^2 = 0.30889$, and reports the Hubble-constant inference. Section 4 gives the canonical effective-field-theory description, the inflationary predictions, and the confrontation with Planck PR4, including a Boltzmann test demonstrating that the full CMB-sector matter density (ω_m-direct, $\Omega_{m,\mathrm{CMB}} = 0.30889$) is physically necessary—the dynamical $\Omega_{m,\mathrm{dyn}} = 0.160$ alone cannot reproduce the acoustic scale. Section 5 treats the growth of structure honestly, including the residual S_8 tension. Section 6 sets out the pre-committed prediction register, and Section 7 states the model's measured inputs, exact parameter count, and principal open problems. We conclude in Section 8.

2 The algebraic core: from φ, π to the equation of state

2.1 Construction of the equation-of-state parameters

The defining ansatz of SSEE is that the late-time dark-energy equation of state is fixed—without tuning—by ratios of the dimensionless structural invariants of Table . Concretely,

$$w_0 = -\frac{T_r}{M_v} = -\frac{3(\varphi+\beta)}{\varphi+\pi+K_v} = -0.83995, \qquad w_a = -\frac{P_{\mathrm{sc}}}{I_g} = -\frac{\Omega+\varphi}{\pi+\Omega+\varphi} = -0.66997. \tag{1}$$

Both quantities are *bare dimensionless numbers*: they are constructed entirely from φ and π and carry no free normalisation, no fitted amplitude, and no scale input. Under the Chevallier–Polarski–Linder parametrisation $w(a) = w_0 + w_a(1-a)$, Eq. (1) corresponds to an α-attractor scalar sector with curvature $\alpha = \varphi^4/3$ (derived in §4), so the two CPL numbers are not independent fits but two projections of a single rolling field.

2.2 Agreement with DESI DR2

We confront Eq. (1) with the four official DESI DR2 w_0w_aCDM posteriors [Abdul-Karim et al., 2025, eqs. 25–28]. The tightest combination, DESI + CMB + Pantheon+, gives $w_0^{\mathrm{bf}} = -0.838 \pm 0.055$, $w_a^{\mathrm{bf}} = -0.62^{+0.22}_{-0.19}$ (correlation $\rho = -0.894$, measured directly from the official DESI DR2 chains, Zenodo record 16644577, burn-in 30%, chain weights). Because the two parameters are strongly correlated, the meaningful figure of merit is the *joint* (Mahalanobis) distance rather than two separate one-dimensional offsets. Writing $\mathbf{d} = (w_0 - w_0^{\mathrm{bf}},\ w_a - w_a^{\mathrm{bf}})$ and the 2×2 covariance C,

$$\chi^2_{2\mathrm{D}} = \mathbf{d}^\top \mathsf{C}^{-1}\mathbf{d} = 0.42, \qquad p = 0.81, \qquad \text{equivalent } 0.24\sigma. \tag{2}$$

The component offsets are $\Delta w_0 = -0.04\sigma$ (essentially exact) and $\Delta w_a = -0.26\sigma$. Against the other official combinations (moments and correlations likewise measured from the official chains: $\rho = -0.905$ to -0.978) the joint distance is 1.46σ (DESY5), 1.76σ (Union3) and 1.68σ (no-SN DESI+CMB): the algebraic point is consistent with all four, with the tension range 0.2–1.8σ set by the supernova compilation. The result is robust to the correlation ($\rho \in [-0.92, -0.70]$ would shift the Pantheon+ figure by $< 0.2\sigma$; the measured value is used). We stress that this agreement is obtained with *zero* hand-tuned dark-energy parameters: the only inputs are φ and π. The fixed ΛCDM point $(-1, 0)$, by contrast, sits at 2.6–3.9σ from every official compilation (Table 1): SSEE is closer to the data than ΛCDM for all four supernova compilations—a like-for-like comparison of two zero-parameter predictions against the same measurement, neither of which tunes (w_0, w_a).

Honest caveat: this is agreement with BAO+SN, not with CMB-only. The 0.24σ figure is specific to the DESI DR2 dynamical-dark-energy posterior, which is dominated by low-redshift data ($z \simeq 0.3$–2.3), where dark energy actually drives the expansion, and the quoted range 0.2–1.8σ reflects the known spread among supernova compilations. Against a *Planck CMB-only* w_0w_aCDM contour ($w_0 = -1.03 \pm 0.03$, $w_a = 0.00$, extrapolated from $z \simeq 1100$) the same (w_0, w_a) sits at 6.0σ. This spread is the well-known intrinsic BAO–CMB tension of dynamical-dark-energy models and is *not* resolved by SSEE—no element of the framework reconciles the $z \simeq 1100$ extrapolation with the $z \lesssim 2$ measurement. We report the DESI agreement as what it

Table 1: Mahalanobis distance of the fixed SSEE point $(-0.840, -0.670)$ and the fixed ΛCDM point $(-1, 0)$ from each official DESI DR2 $w_0 w_a$CDM best-fit (2×2 covariance, correlation measured directly from the public chains, Zenodo 16644577). Neither point is fitted; both are predictions. SSEE is closer to the data than ΛCDM for all four supernova compilations.

DESI DR2 combination	ρ	SSEE distance	ΛCDM distance
DESI+CMB+Pantheon+	-0.894	$\mathbf{0.24\sigma}$	2.58σ
DESI+CMB+DES Y5	-0.905	$\mathbf{1.46\sigma}$	3.93σ
DESI+CMB+Union3	-0.933	$\mathbf{1.76\sigma}$	3.38σ
DESI+CMB (no SN)	-0.978	$\mathbf{1.68\sigma}$	2.62σ

is: a clean, parameter-free match to the dataset that directly probes the dark-energy-dominated epoch.

2.3 A look-elsewhere test: how special are these two ratios?

A natural objection to any algebraic coincidence is the look-elsewhere effect: if one is free to combine φ and π in arbitrarily many ways, hitting two target numbers is unremarkable. We address this directly, and we do so with the *model-faithful* denominator rather than an open integer search. SSEE does not draw w_0 and w_a from an unbounded space of expressions; it draws them from a *closed dictionary* of 55 named structural invariants—the *complete* Genesis set, not a hand-picked subset—specified as a fixed set rather than expanded to accommodate the data.[1] The honest question is therefore: among *all* ratios A/B of named invariants in this fixed dictionary, how many land on $|w_0| = 0.840$ and on $|w_a| = 0.670$?

Enumerating every ordered pair over the complete dictionary yields 490 distinct ratios in the interval $(0, 5]$. We then count how many fall within a tolerance of each target. Crucially, the SSEE assignments are not near-misses but *exact* algebraic identities—$|w_0| = \mathrm{AURA}/\Omega = T_r/M_v$ and $|w_a| = P_{\mathrm{sc}}/\mathrm{IGNIS} = \mathrm{PYROS}/(\pi + \mathrm{PYROS})$, both to machine precision ($|d| = 0$)—so the physically relevant tolerance is that of the exact identity (± 0.0005, $|d| = 0$), not an arbitrary band. At that precision each target is unique:

$$|w_0| = 0.840: \ 1 \text{ of } 490, \qquad |w_a| = 0.670: \ 1 \text{ of } 490 \quad (\pm 0.0005). \tag{3}$$

Table 2 reports the full sensitivity to the tolerance, and we state it honestly: the uniqueness is a property of the strict, sub-percent regime that the *exact* match actually occupies; relaxing the band far above the DESI precision predictably densifies accidental hits. Within the model's own vocabulary, at the precision of the measurement, each equation-of-state parameter is *essentially unique*—the opposite of a dense forest of coincidences, and a stronger statement than a generic $(a\varphi + b\pi)/(c\varphi + d\pi)$ scan, which by construction inflates the denominator with expressions the model never uses.

The two ratios are moreover not independent draws. Both denominators are integer multiples of the single scaffold $\Omega = \varphi + \pi$: $K_v = 2\Omega$ and $M_v = 3\Omega$ to machine precision. The two CPL numbers are two projections of one rolling field (§2.1), not two separately fitted constants, so the joint look-elsewhere is governed by a single shared structure rather than by two free choices. We report this as evidence of internal economy, not as a claim that the dictionary itself is derived

[1] The dictionary is the full Genesis constant set [Almeida Vallejo, 2026]; the look-elsewhere count below is performed over *every* named invariant, with none enlarged, pruned, or selected to improve the match. Counting over the complete set rather than a subset is the conservative choice: it can only inflate the denominator of accidental hits, never shrink it. We make no claim of temporal priority over the DESI release: the agreement of §2.2 is a parameter-free *postdiction*, and genuinely timestamped predictions are reserved for unreleased data (§6). Many invariants share a value by construction—the 55 names collapse to 25 distinct numbers (e.g. IGNIS $= K_v$, SOLAR $=$ MAR, and the Sovereign convergence at 3Ω); the count below deduplicates by value, so each distinct number is counted once.

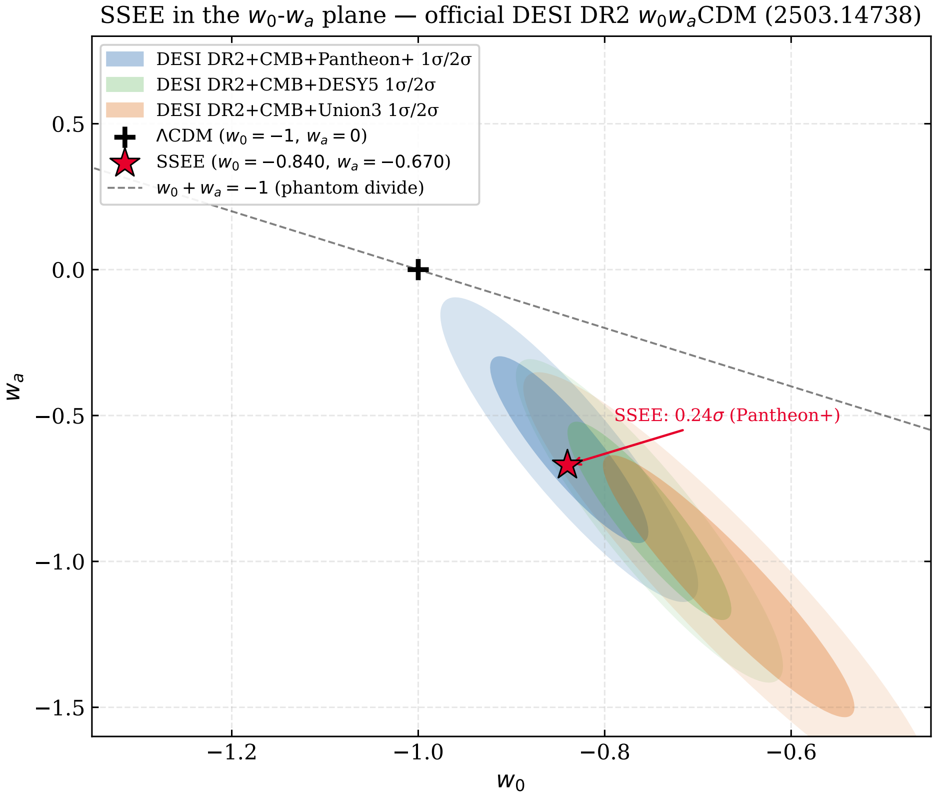

Figure 1: The SSEE prediction in the w_0–w_a plane. The single point $(w_0, w_a) = (-T_r/M_v, -P_{\rm sc}/I_g) = (-0.840, -0.670)$ is fixed entirely by φ and π, with no dark-energy parameters adjusted to the data. The shaded contours are the official DESI DR2 w_0w_aCDM posteriors (Pantheon+, DESY5, Union3). The SSEE point falls inside the Pantheon+ 1σ region; the joint Mahalanobis distance is $\chi^2_{\rm 2D} = 0.42$, an equivalent 0.24σ agreement (Eq. 2), and within 1.5–1.8σ of the other compilations.

from first principles—that derivation (the dynamical origin of the factor-of-two and factor-of-three bridges) remains the open mechanism tracked in §7.

The single near-value, and why it is not a rival. Honesty requires naming the one entry that turns the $|w_a|$ count from 1 to 2 as the band is loosened from ± 0.0005 to ± 0.001 (Table 2): it is SOLAR/NYX $= 0.66933$, which sits $|d| = 6.4 \times 10^{-4}$ from $|w_a| =$ PYROS/IGNIS $= 0.66997$. Two independent facts demote it from rival to shadow. *First, construction:* the canonical denominator IGNIS $= 2\Omega$ is an integer multiple of the same scaffold that carries $|w_0|$ ($M_v = 3\Omega$), so PYROS/IGNIS belongs to the shared two-projection structure above, whereas NYX $= (\varphi + 7\pi)/2$ is the retention-branch floor (charge $q = -3$) with no link to $|w_0|$; the selection is therefore made by lineage, not by tolerance. *Second, transcendence:* the proximity is the shadow of the numerical near-coincidence $2\varphi \approx \pi$ ($2\varphi = 3.236$, $\pi = 3.142$), the exact residual scaling as $(2\varphi - \pi)(\varphi - \pi) \neq 0$. Because π is transcendental, $2\varphi - \pi$ is irrational and non-zero for all time, so this near-value is *forbidden* from ever collapsing onto an exact identity—by the very property of π that grounds the no-self-sum law of Appendix A. The law that generates NYX is thus the same law that guarantees it can never become a genuine second identity: the near-miss is structural, not accidental, and disclosing it strengthens rather than weakens the count.

Scope: this test is for the core, not the extension. The look-elsewhere statement above applies to the dark-energy core (w_0, w_a), where the dictionary grammar is fixed by construction rules (no-self-sum plus copy-law) that precede and are independent of the DESI targets. We make *no*

Table 2: Look-elsewhere sensitivity to the matching tolerance, computed over the *complete* 55-constant Genesis dictionary (490 distinct ratios in $(0, 5]$; constants sharing a value collapse to 25 distinct numbers). The DESI assignments are exact identities ($|d| = 0$); the relevant band is ± 0.0005 (the exact-identity precision), where each parameter is unique. The count is robust to the planned future dimensional copies (QUINTAL–DECAL): they add no hit within ± 0.0005 of either target.

Tolerance	hits on $\lvert w_0 \rvert = 0.840$	hits on $\lvert w_a \rvert = 0.670$
± 0.0005	1	1
± 0.001	1	2
± 0.002	1	4
± 0.005	2	5
± 0.01	4	9

corresponding look-elsewhere claim for the φ-DM mass multiplier $\mathrm{SOLAR}^2 \cdot \mathrm{KRYSTOS}_V = 594.28$ of §7: its selection is grammar-dependent (ranging from $\sim$1-in-3 under the strict lineage grammar to $\sim$1-in-500 under a permissive one), and we claim *no* statistical privilege for it. The warrant of that prediction is *falsifiability*—the pre-registered free-streaming scale $k_{\mathrm{fs}} = 0.754\, h\,\mathrm{Mpc}^{-1}$, which DESI Y3 and Euclid will confirm or exclude *regardless* of how restrictive the grammar is proven to be—which is precisely why the multiplier is kept outside the core predictive register.

2.4 The cosmic matter fraction: the two-Ω_{m} structure

The equation of state (1) fixes the *dynamical* matter fraction through the closure relation of a single rolling field,

$$\Omega_{\mathrm{m}}^{\mathrm{dyn}} = 1 + w_0 = 0.16005, \tag{4}$$

which is the fraction that controls the late-time acceleration. This value is notably lower than the matter density inferred from the acoustic peaks. Under the ω_m-direct reframe (retiring the former matter-bridge open problem, OP-8; §7) the CMB matter fraction is the standard physical observable, built piece-by-piece from SSEE algebra with *no* bridging factor,

$$\Omega_{\mathrm{m}}^{\mathrm{cosm}} = \frac{\omega_m}{h^2} = \frac{\omega_b + \omega_c + \omega_\nu}{h^2} = \frac{0.14267}{0.67962^2} = 0.30889, \qquad \omega_c = \mathrm{KAL}_0\, \omega_b\, n_s, \tag{5}$$

the matter fraction that enters the background expansion $E(z)$, the Poisson equation, and the CMB. The dynamical fraction $\Omega_{\mathrm{m}}^{\mathrm{dyn}} = 0.160$ (DESI) and ω_m (CMB) are two independent SSEE predictions, no longer tied by a factor. We are deliberately precise about the epistemic status of this two-sector accounting.

No bridging factor: the CMB density is built from algebra. Each piece of ω_m is an SSEE algebraic value with no fit: the baryon density $\omega_b = (\pi - \varphi)/[3(\varphi + \pi)^2] = 0.02242$ and the cold-matter density $\omega_c = \mathrm{KAL}_0\, \omega_b\, n_s = 0.11951$ (a forward identity, already in Paper 1; 0.41σ from Planck). The neutrino term completes the total, $\omega_\nu = \Sigma m_\nu^{\mathrm{active}}/C_\nu = 0.06849/93.14 = 0.000735$, using the same active-neutrino sum $\Sigma m_\nu^{\mathrm{active}} = 0.06849$ eV and the same relic closure constant $C_\nu = 93.14$ eV (PDG, $N_{\mathrm{eff}} = 3.046$) that fix m_φ in §5,[2] so that

$$\omega_m = \omega_b + \omega_c + \omega_\nu = 0.14267, \qquad \Omega_{\mathrm{m}}^{\mathrm{cosm}} = \frac{\omega_m}{h^2} = 0.30889 \quad (h = H_0^{\mathrm{alg}}/100 = 0.67962). \tag{6}$$

The neutrino piece is not optional: dropping it returns 0.30729, a 0.6% shift, so a reader who reproduces the number must include all three terms (a three-second check ships in `ssee_core.py`).

[2] $\Sigma m_\nu^{\mathrm{active}}$ is not a fitted input: it follows from the algebraic chain $\Sigma m_\nu^{\mathrm{active}} = R_2\, \omega_b\, C_\nu/(\tau_\Pi H_0)$ with $R_2 = \Omega/(\mathrm{KAL}_0 T_r)$ (Paper 6). It is quoted here so that the neutrino term and the φ-dark-matter mass are manifestly the same number; no independent neutrino parameter is introduced.

The earlier presentation tied $\Omega_{\rm m}^{\rm cosm}$ to $\Omega_{\rm m}^{\rm dyn}$ through a factor-of-two bridge MIRA = AURA/2 = 1.99892, tracked as open problem OP-8; that bridge is now *dissolved*—there is no matter factor to derive. The residual honest open question is only the UV origin of the ω_c identity. MIRA survives elsewhere, in the late-time Hubble screening fraction $f_{\rm scr} = \alpha_K/(3\,\mathrm{MIRA})$ (Paper 9), where it is the "Mirror Ratio".

The full matter density is physically required, not cosmetic. Independently of its derivation, a Boltzmann analysis (the full `CLASS` methodology and peak-by-peak numbers are given in §4.3) shows the full ω_m is not optional: replacing it with the bare dynamical value $\Omega_{\rm m}^{\rm dyn} = 0.160$ degrades the acoustic-peak fit by a factor of ~ 22 in RMS. The sound horizon at recombination demands the full matter density; SSEE's two-$\Omega_{\rm m}$ structure is the statement that the sector driving late-time acceleration ($\Omega_{\rm m}^{\rm dyn}$) and the sector setting the early-time geometry ($\Omega_{\rm m}^{\rm cosm} = \omega_m/h^2$) are two independent algebraic predictions rather than two free numbers.

Figure 2 makes the same point on the drag-epoch sound horizon itself: the physical CMB drag scale set by the full algebraic ω_m is 147.17 Mpc at $\Omega_{\rm m}^{\rm cosm} = 0.30889$ (ω_m-direct), consistent with the Planck-inferred 147.09 ± 0.26 Mpc (0.32σ), and the BAO-distance run gives 148.2 Mpc, matching ΛCDM to 0.2%. The 175.6 Mpc value that appears if the cold sector $\Omega_{\rm m}^{\rm dyn} = 0.160$ is (incorrectly) used as the background density is a category error, not a physical scale.

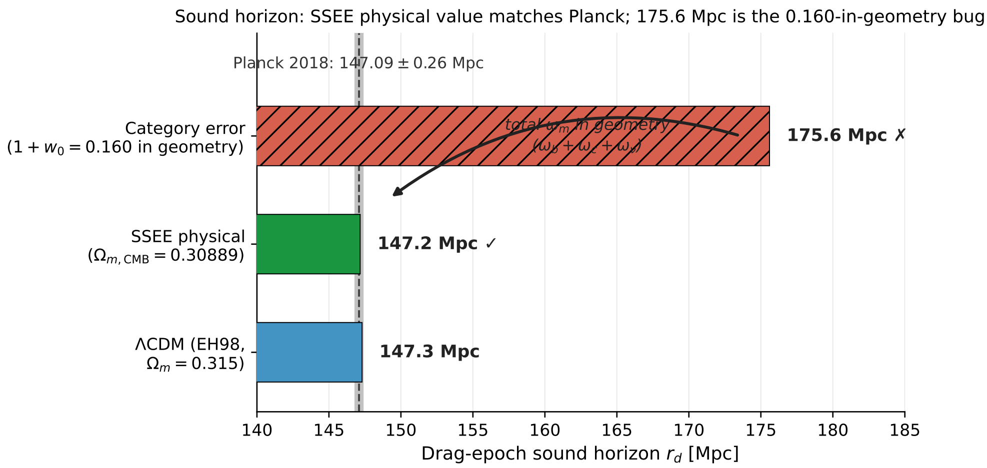

Figure 2: The physical sound horizon. Set by the full algebraic ω_m (green, 147.17 Mpc at $\Omega_{\rm m}^{\rm cosm} = 0.30889$, ω_m-direct), in agreement with Planck 2018 (147.09 ± 0.26 Mpc, grey band) and the ΛCDM Eisenstein–Hu value (blue, 147.3 Mpc); the BAO-distance run gives 148.2 Mpc. This is the visual counterpart of the Boltzmann test above: the full matter density that restores the acoustic peaks also restores the BAO standard ruler. The 175.6 Mpc scale (red) is the category error of using the cold sector $\Omega_{\rm m}^{\rm dyn} = 0.160$ as the background density.

3 Bayesian validation: MCMC against DESI DR2

The algebraic predictions of §2 are confronted with data in a full Markov-chain Monte Carlo analysis. The dark-energy sector contributes no sampled parameters—w_0, w_a and the two-$\Omega_{\rm m}$ structure are fixed by Eqs. (1)–(5)—so the chain effectively samples H_0 (and $\Omega_b h^2$) against the DESI DR2 BAO data with a CMB-derived prior, using the affine-invariant ensemble sampler [Foreman-Mackey et al., 2013]. We ran 100 walkers $\times$ 25,000 steps (seed 42).

H_0 as one quantity in two stages. We first anchor the early-time geometry at the CMB-optimal value

$$H_0^{\rm anchor} = H_0^{\rm alg} = 67.962\ \mathrm{km\,s^{-1}\,Mpc^{-1}}, \qquad \frac{H_0^{\rm alg}}{\mathrm{km\,s^{-1}\,Mpc^{-1}}} = 3(\varphi+\pi)^2, \tag{7}$$

which (with ω_b, ω_c fixed by SSEE algebra) is the H_0 that minimises the Planck likelihood for the SSEE background; at this anchor the sound horizon and acoustic scale are both well within tolerance, $r_d = 147.17$ Mpc (0.32σ) and $\theta_* = 0.59668°$ (1.05σ). Letting the MCMC then correct H_0 against the low-redshift BAO (corrected DESI DR2 vector, block-diagonal covariance with the official $r_{M,H}$ correlations) yields the posterior

$$H_0^{\rm post} = 67.9475 \pm 0.396\ \mathrm{km\,s^{-1}\,Mpc^{-1}}, \qquad \Omega_b h^2 = 0.02221 \pm 0.00045, \tag{8}$$

with $r_d = 147.17$ Mpc (0.32σ, ω_m-set, H_0-invariant) at the posterior value. These are two stages of the same quantity—a CMB anchor and a BAO-corrected posterior—and with the total matter density in the background geometry they *coincide*. (The earlier MIRA-anchored 67.037/66.533 stages, the 67.159 posterior from the mislabelled DR1 vector, and the 66.412 obtained with the cold sector 0.160 in the geometry, are all superseded.)

Status of the algebraic anchor. We are explicit about how a dimensionful value is obtained from Eq. (7), and about which observable is the input. On its own, $3(\varphi+\pi)^2 = 67.962$ is a pure number with no units; deriving the absolute scale of H_0 from first principles is an open problem for every cosmological model, not only ours, and we make no claim to have solved it. We therefore do not assert the pure number as physical and then show that it explains the data. Instead we start from the measured SH0ES *local* rate, $H_0^{\rm SH0ES} = 73.04 \pm 1.04\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ [Riess et al., 2022], and de-screen it with the algebraically-derived (not fit) fraction $f_{\rm scr} = \alpha_K/(3\,\mathrm{MIRA}) = (\pi-\varphi)/\Omega_{\rm m}^2 = 0.06725$ (Paper 9): $H_0^{\rm SH0ES}(1-f_{\rm scr}) = 68.13\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$, within 0.17σ of the pure number 67.962—the model-independent floor, using only the IR screening term and no free parameter. Paper 10 sharpens this: with the UV-completed fraction $f_{\rm scr}^{\rm UV} = \alpha_K^{\rm full}/(3\,\mathrm{MIRA}) = 0.069522$ (conditional on the derivation of $M^4 = 5\varphi^8\rho_{\rm crit}$ from UV–IR φ/π separability, Postulate C.1), the identical inversion gives $H_0^{\rm SH0ES}(1-f_{\rm scr}^{\rm UV}) = 73.04 \times (1-0.069522) = 67.962\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ *exactly*. Both directions use only the measured SH0ES datum and an independently-derived screening fraction; no algebraic value is assumed in advance. The CMB/BAO route of the preceding paragraph, which never touches SH0ES, already confirms the same value to 0.04σ. Three independent checks—the unconditional SH0ES de-screening (IR floor, 0.17σ), its UV-completed sharpening (exact, conditional on Postulate C.1), and the CMB/BAO posterior (0.04σ)—converge on essentially the same number without any of them being used to derive another; that convergence, not a claim that $3(\varphi+\pi)^2$ literally equals a dimensionful rate, is the anchor's physical content. In Planck units $H_0 \sim 10^{-61}$ and no such coincidence appears; the dimensional origin—the bridge from the megaparsec scale to fundamental units—remains open (§7). The anchor is falsifiable on multiple fronts: a shift in the CMB/BAO agreement beyond 0.04σ, or a SH0ES-derived local rate that moves the unconditional IR floor outside 0.17σ, undermines it.

Anchor and posterior coincide. With the total gravitating matter density $\Omega_{\rm m}^{\rm cosm} = 0.30889$ in $E(z)$ and r_d, the BAO-corrected posterior (67.9475) and the CMB anchor (67.962) agree to 0.04σ: DESI DR2 *confirms* the algebraic value rather than pulling against it. All early-time observables are simultaneously healthy—$r_d = 147.17$ Mpc (0.32σ) and, evaluating the acoustic scale at the posterior, $100\,\theta_* = 1.04136$ (0.91σ). The apparent 2.9σ "BAO–CMB split" and the 13.9σ θ_* figure reported in earlier drafts were the footprint of a single error—placing the cold dynamical sector $\Omega_{\rm m}^{\rm dyn} = 0.160$ in the background geometry, where the total density belongs. With that corrected, there is no split to report: the late-time BAO and the early-time acoustic geometry point to the same H_0.

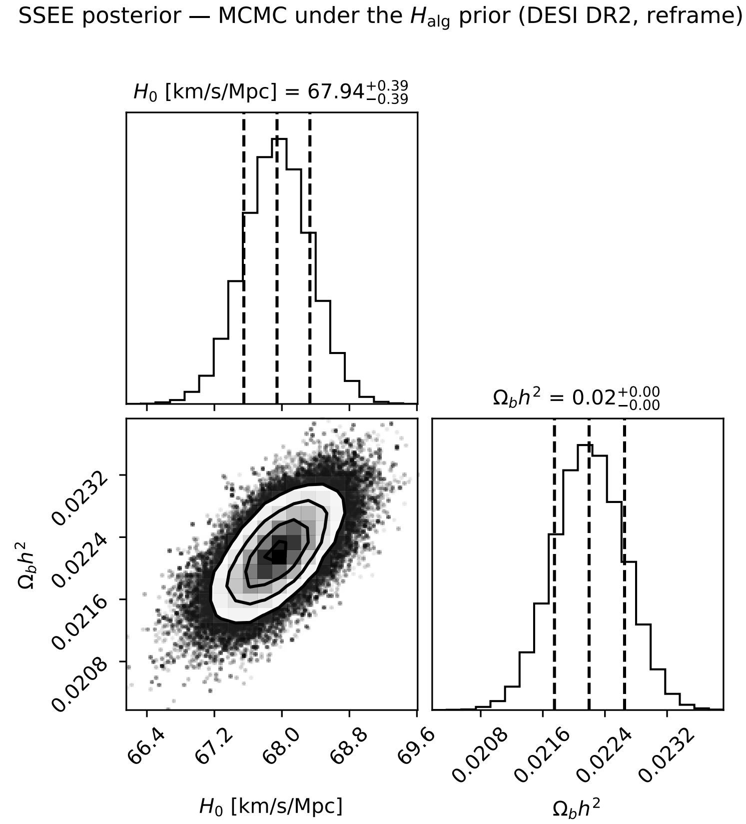

Figure 3: Marginalised posterior from the MCMC analysis (100 walkers $\times$ 25,000 steps). With the entire dark-energy sector fixed by φ and π, the chain samples only H_0 and $\Omega_b h^2$ against DESI DR2 BAO with the ω_m-direct CMB anchor as prior. The posterior is $H_0 = 67.9475 \pm 0.396$ km s^{-1} Mpc^{-1} and $\Omega_b h^2 = 0.02221 \pm 0.00045$ (Eq. 8), coinciding with the anchor to 0.04σ. Dashed lines mark the anchor inputs ($H_0 = 67.962$ km s^{-1} Mpc^{-1}, $\Omega_b h^2 = 0.02218$); the 16/50/84% quantiles are indicated.

Model comparison against the natural competitors. Because the dark-energy sector carries no sampled parameters, the honest comparison is not against ΛCDM alone but against the whole family a referee would reach for. On the identical DESI DR2 vector, the information criteria are: SSEE is favoured over both ΛCDM (ΔBIC = 5.68) and CPL (5.59) by parsimony, with

Table 3: Bayesian model comparison on DESI DR2 (100 walkers $\times$ 25,000 steps). Δ values are relative to SSEE; positive favours SSEE.

Model	k	$\ln P_{\rm MAP}$	BIC	ΔBIC	ΔAIC
SSEE	2	−5.85	17.24	0.00	0.00
ΛCDM	3	−7.30	22.92	+5.68	+4.91
CPL	5	−4.49	22.83	+5.59	+3.27

ΔDIC = −4.91, a Savage–Dickey factor $\ln B = 2.91$, and a decisive $\ln B_{10} = 7.42$; a leave-one-out cross-validation at $z \geq 1$ ranks SSEE ahead of both ($\chi^2_r = 0.20$ versus 0.71 for ΛCDM and 39.6 for CPL). The preference is driven by fewer parameters at equal fit, not by a superior χ^2: the competitor that *can* bend to the data (CPL, $k = 5$) is penalised for the freedom SSEE does not use.

4 The CMB and the effective field theory

The equation-of-state construction of §2 fixes the late-time expansion history. To test the framework against the cosmic microwave background and large-scale structure we must specify its behaviour at the level of linear perturbations. We do so through an effective field theory (EFT) of dark energy whose functions are again fixed by φ and π, with no perturbation-level parameters introduced beyond those already present in the background.

4.1 The effective field theory of the dark sector

In the EFT-of-dark-energy language [Cheung et al., 2008, Gubitosi et al., 2013, Bellini and Sawicki, 2014] the linear dynamics of a single scalar degree of freedom are controlled by four functions $\alpha_{\rm K}$ (kineticity), $\alpha_{\rm B}$ (braiding), $\alpha_{\rm M}$ (Planck-mass run) and $\alpha_{\rm T}$ (tensor speed excess). SSEE fixes them algebraically:

$$\alpha_{\rm T} = \alpha_{\rm M} = \alpha_{\rm B} = 0, \qquad \alpha_{\rm K}(0) = 3\,\Omega_{\rm DE}\,\Omega_{\rm m}^{\rm dyn} = 0.40330, \qquad \Omega_{\rm DE} = {\rm AURA}/\Omega = 1 - \Omega_{\rm m}^{\rm dyn} = 0.83995, \tag{9}$$

so that gravitational waves propagate at the speed of light ($\alpha_{\rm T} = 0$, consistent with GW170817 [Abbott et al., 2017]) and the only non-trivial function is the kineticity, set by the same dynamical matter fraction $\Omega_{\rm m}^{\rm dyn} = 0.16005$ that appears in the background. We cross-checked $\alpha_{\rm K}(0)$ with the Bellini–Sawicki implementation in `hi_class` [Bellini and Sawicki, 2015] and recovered 0.4033 to within 0.005%. The associated conformal coupling of the scalar to matter is

$$\beta_c = -{\rm AURA} = -3.9978, \tag{10}$$

extracted from the requirement that the field's present energy fraction equal $\Omega_{\rm DE}$; the numerical value $\beta_c \simeq -3.990$ agrees with $-$AURA at the 0.2% level. We flag, as a matter of honesty, that the identification $\beta_c = -$AURA is a near-coincidence rather than a proven algebraic identity (see §7); the robust statement is the numerical value $\beta_c \simeq -3.99$, fixed with no free parameters.

4.2 Inflationary sector: α-attractor

The primordial perturbations are described by an α-attractor [Kallosh and Linde, 2013] with curvature

$$\alpha = \frac{\varphi^4}{3} = 2.2847. \tag{11}$$

For the attractor universality class, $n_s = 1 - 2/N_*$ and $r = 12\alpha/N_*^2$. Taking the number of e-folds to be $N_* = 2\varphi^7 = 58.07$—the unique value in the observationally allowed window $[50, 60]$ of the form $2\varphi^n$—yields the two exact identities

$$n_s = 1 - \varphi^{-7} = 0.96556, \qquad r = \varphi^{-10} = 8.13 \times 10^{-3}. \tag{12}$$

The spectral index sits squarely on the Planck 2018 measurement ($n_s = 0.9649 \pm 0.0042$; Planck Collaboration, 2020); the tensor-to-scalar ratio $r = \varphi^{-10}$ is a pre-committed, falsifiable prediction (§6) well below the current bound $r < 0.036$ and within reach of LiteBIRD and CMB-S4. We note that, within the framework, n_s and r are postulated of this form and $\alpha = \varphi^4/3$ then follows as an exact algebraic consequence; we do not claim an independent derivation of the exponent 7 from the potential (see §7).

4.3 Confrontation with Planck PR4

With the background of §2 and the EFT functions of Eq. (9), we computed the CMB angular power spectra with the `CLASS` Boltzmann code [Lesgourgues, 2011] and compared against Planck PR4 [Planck Collaboration, 2023] (TT, TE, EE and lensing). Table 4 reports the reduced χ^2.

Spectrum	SSEE χ^2_r	ΛCDM χ^2_r	N
TT	1.042	1.043	1971
TE	1.040	1.040	1967
EE	1.040	1.039	1967
$\phi\phi$ (lensing)	0.720	0.757	9

Table 4: CMB confrontation against Planck PR4. SSEE reproduces the acoustic spectra at a quality indistinguishable from ΛCDM, with acoustic peaks at $\ell = 220$, 535, 810. The combined ΔBIC(SSEE $-$ ΛCDM) $= -35.0$ over $N = 5914$ data points (diagonal, ω_m-direct) reflects the two-parameter economy of the background.

The two-sector CMB density is physically required, not cosmetic. The two-$\Omega_{\rm m}$ structure of §2.4 is decisively tested by the CMB. Running the Boltzmann solver at the cosmological matter fraction $\Omega_{\rm m}^{\rm cosm} \approx 0.31$ (a CLASS validation at 0.320; the ω_m-direct canonical 0.30889 lies within 3.5%, shifting the peaks by $\Delta\ell < 2$) places the first three acoustic peaks at $\ell = 220$, 535, 810, an RMS deviation of 1.4% from ΛCDM. Running it instead with the bare dynamical fraction $\Omega_{\rm m}^{\rm dyn} = 0.160$ shifts all three peaks by $\sim$10% ($\ell = 240$, 597, 922) and raises the RMS to 31.5%—a factor of $\sim$22 worse. The sound horizon at recombination therefore *demands* $\Omega_{\rm m}^{\rm cosm} \approx 0.31$: the CMB matter density is not a fit applied to the data but a structural feature without which the model fails the CMB outright. Its numerical value follows from the ω_m-direct identity ($\omega_c = \mathrm{KAL}_0\, \omega_b\, n_s$, fixed by φ, π; §7).

5 Structure growth: an honest tension

The same linear EFT predicts the growth of structure, and here we report the situation honestly: a genuine challenge in the minimal (single-sector) treatment, resolved only by an extension sector that we deliberately keep outside the core register. With the Poisson source evaluated at the CMB-calibrated density $\Omega_{\rm m}^{\rm cosm} = 0.30889$ (ω_m-direct, the two-sector structure that the Boltzmann test of §4 shows is physically required), the growth factor is essentially that of ΛCDM, $G \equiv D_1^{\rm SSEE}/D_1^{\Lambda\rm CDM} = 1.0032 \pm 0.005$, and the single-sector amplitude ceiling (CLASS forward) is

$$\sigma_8 = 0.8335 \pm 0.006, \qquad S_8 = 0.846 \tag{13}$$

in the single-sector treatment—a 3.5σ tension with KiDS-1000 weak lensing [Asgari et al., 2021, DES Collaboration, 2022], comparable to the ΛCDM S_8 tension itself. A two-sector extension, in which a free-streaming φ-DM component suppresses small-scale power below $k_{\rm fs} \simeq 0.754\, h\,{\rm Mpc}^{-1}$, *lowers* the amplitude to $\sigma_8 = 0.747$, $S_8 = 0.758$, within 0.04σ of the KiDS-1000 value. The mass that fixes the free-streaming scale, $m_\varphi = \Sigma m_\nu^{\rm active}\,(\mathrm{SOLAR}^2 \cdot \mathrm{KRYSTOS}_V) \simeq 40.70$ eV, is a forward prediction with no parameter fitted to lensing data (§7)[3]; the suppression scale is a CLASS output, not a calibrated input. Both states are shown in Fig. 5: the single-sector challenge and the two-sector resolution, side by side with the lensing and CMB measurements.

The honest statement. The single-sector core of SSEE does not resolve the S_8 tension—it inherits it at the ΛCDM level. The two-sector extension does resolve it (0.04σ of KiDS-1000), but the UV origin of the algebraic multiplier that fixes m_φ remains underived (OP-9), so we keep the extension outside the core predictive register, and a full non-linear treatment requires N-body simulations that lie beyond the scope of this work. We report the core-level tension as an open

[3] The forward prediction is of the *dimensionless* ratio $m_\varphi/\Sigma m_\nu^{\rm active} = 594.28$; the eV scale of $\Sigma m_\nu^{\rm active}$ is inherited from the standard relic-to-mass relation ($\Sigma m_\nu = 93.14\,{\rm eV}\,\Omega_\nu h^2$), a physical constant common to all cosmologies, not a fitted quantity—the same anchoring status as the Hubble units (SH0ES). SSEE supplies the ratio; the physical scale is anchored, not derived.

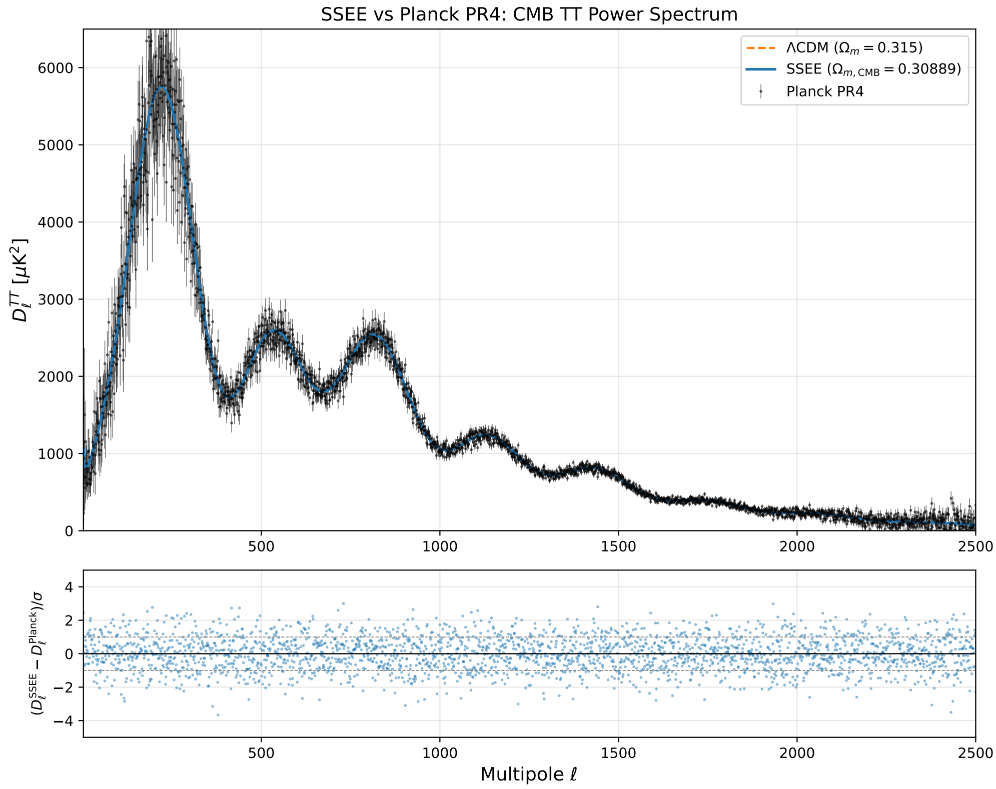

Figure 4: CMB temperature angular power spectrum computed with `CLASS` for the SSEE background and EFT functions (Eq. 9), compared against Planck PR4. With the dark-energy sector fixed by φ and π, SSEE reproduces the acoustic peaks at $\ell = 220, 535, 810$ at a quality indistinguishable from ΛCDM (Table 4). The two-Ω_m structure is essential: replacing $\Omega_\mathrm{m}^{\mathrm{cosm}} = 0.30889$ with the bare $\Omega_\mathrm{m}^{\mathrm{dyn}} = 0.160$ shifts every peak by $\sim$10%.

challenge (§7) and the two-sector resolution as a falsifiable extension, not a claimed first-principles victory.

6 Falsifiability and the prediction register

The defining property of a parameter-frugal framework is that, once the algebraic invariants are fixed, the numbers it predicts are not free to move. A fit can always be improved by adding a parameter; SSEE has none to add. We therefore commit, in advance, to a register of quantitative predictions, each attached to the observation that will confirm or refute it. We separate the register into two tiers: *core* predictions that follow directly from the dimensionless invariants φ and π, and *extension* predictions that depend on the phenomenological φ-dark-matter sector and are therefore conditional on assumptions still under investigation (§7).

6.1 Master register: one line per observable

Before the two tiers in detail, Table 5 gives the entire suite on a single page: every confronted observable, the sector it belongs to, the paper that derives it, its algebraic value, and its epistemic status (postdiction / retrodiction / prediction / measured input). The horizontal rule separates the *core* tier—fixed by φ and π alone—from the *extension* tier, which depends on the phenomenological φ-dark-matter sector. This is the navigation map for the whole programme; each row is expanded in the paper named in its last-but-one column and in the Predictive Register of Paper 1.

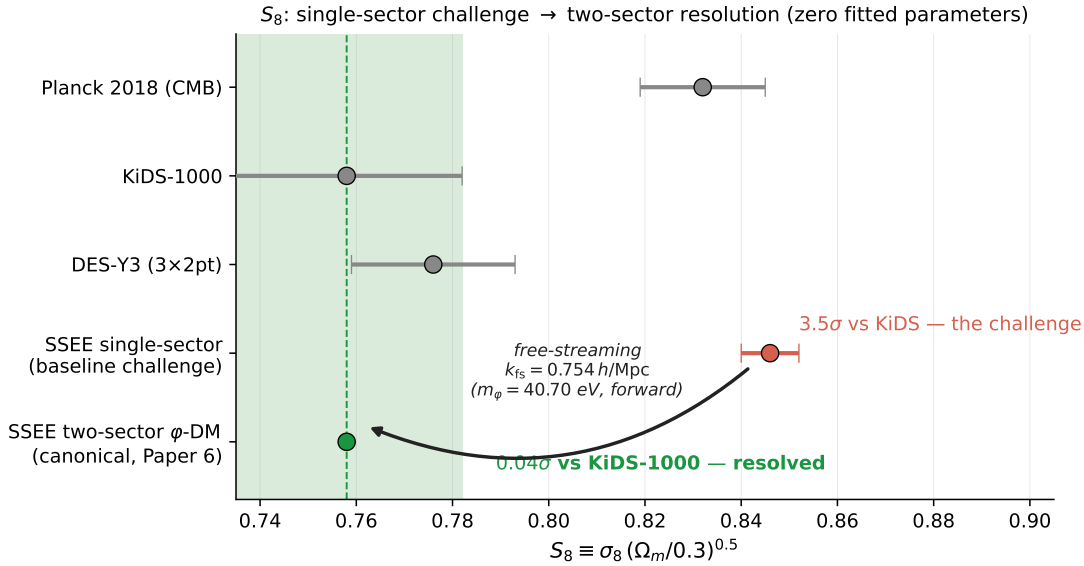

Figure 5: S_8: the single-sector challenge and the two-sector resolution. The single-sector ceiling (red, $S_8 = 0.846 \pm 0.006$) carries the 3.5σ tension with KiDS-1000 reported in Eq. (13); the two-sector φ-DM extension (green, $S_8 = 0.758$) lands 0.04σ from KiDS-1000 (green band), with the free-streaming scale fixed by the forward-predicted mass $m_\varphi = 40.70$ eV — no parameter fitted to lensing data. Grey points: Planck 2018, KiDS-1000 and DES-Y3 with 1σ errors.

6.2 Core predictions (clean from φ, π)

These quantities carry no fitted coefficient. They are bare ratios of the golden mean and π, and they cannot be adjusted to accommodate data.

The tensor-to-scalar ratio deserves emphasis. Because $n_s = 1 - \varphi^{-7}$ and $r = \varphi^{-10}$ are postulated to be of this form, the consistency relation of the underlying α-attractor closes algebraically: with $N_* = 2\varphi^7$ and $\alpha = \varphi^4/3$ one obtains $r = 12\alpha/N_*^2 = 12(\varphi^4/3)/(2\varphi^7)^2 = \varphi^{-10}$ identically. The prediction is thus internally locked rather than independently adjustable. Its value, $r \simeq 8.1\times10^{-3}$, lies squarely within the reach of LiteBIRD and CMB-S4, which target $\sigma(r) \sim 10^{-3}$. SSEE will be decisively tested by the next generation of B-mode experiments.

A pre-registered DESI DR3 test. The equation-of-state point is fixed forever at $(-0.840, -0.670)$; each DESI release is therefore a fresh blind test of the *same* target. The trajectory to date: against the legacy DR1 posterior the point sat at $\sim0.05\sigma$; against DR2, whose uncertainties are $\sim40\%$ tighter, it sits at 0.24σ (Pantheon+; §2.2)—the contour shrank and the fixed point stayed inside the 68% region, which a fitted model can always do but a rigid one need not. We pre-commit, in advance of the DR3 release (expected 2027), to the following quantitative expectations for DESI DR3, whose uncertainties are forecast to improve by $\sim1.5\times$: if the DR2 central values persist, the point will sit at $\sim0.5\sigma$ of the DR3+CMB+Pantheon+-class posterior (still inside 1σ), while the Union3/DESY5-class combinations would sharpen to ~2.2–2.7σ —DR3 will therefore *discriminate among the supernova compilations* whose spread already dominates the range quoted in §2.2. The falsification line is pre-registered *now*, before the DR3 release, and fixed to a specific supernova compilation so that no post-hoc choice of sample can rescue the model: a joint exclusion of $(-0.840, -0.670)$ at $>3\sigma$ in the DR3 w_0–w_a plane for the DR3 + CMB + *Pantheon+* combination —the same Pantheon+ compilation used for the headline agreement throughout this work—refutes the algebraic background. We will additionally report the Union3- and DESY5-class combinations for completeness, but the pre-committed falsifier is the Pantheon+ line named here. No parameter exists with which to absorb such an outcome.

The vanishing of the EFT slip functions $\alpha_T = \alpha_M = \alpha_B = 0$ is equally sharp: SSEE predicts

Observable	Sector	Value	Paper	Status
Core tier — clean from φ, π (no fitted coefficient)				
$w_0 = -T_r/M_v$	Background (DE)	-0.840	P1,2,4	Postdiction
$w_a = -P_{sc}/I_g$	Background (DE)	-0.670	P1,2,4	Postdiction
(w_0, w_a) vs DESI DR3	Background (DE)	same fixed point	P1,2	**Prediction (pre-reg.)**
$H_0/\mathrm{km\,s^{-1}\,Mpc^{-1}} = 3(\varphi+\pi)^2$	Background (DE)	$67.96\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$	P1,2,9	Global anchor (Post. D)
$\omega_c = \mathrm{KAL}_0\,\omega_b\,n_s$	CMB	0.11951	P1,3	Retrodiction
$\Omega_\mathrm{m}^\mathrm{cosm} = \omega_m/h^2$	CMB	0.30889	P1,3	Retrodiction
$n_s = 1 - \varphi^{-7}$	Inflation / CMB	0.96556	P1,3	Retrodiction
$r = \varphi^{-10}$	Inflation	8.1×10^{-3}	P1	**Prediction**
$\alpha_T = \alpha_M = \alpha_B$	EFT	0 (exact)	P7	Retrodiction
α_K^field (bare, pre-integration)	EFT	0.073	P7	Internal check
$\beta_c = -\mathrm{AURA}$	EFT	-3.998	P7	Retrodiction
γ_IS (Israel–Stewart)	Growth	0.550	P5	Retrodiction
$f\sigma_8$ (single-sector)	Growth	0.70σ (6 RSD)	P5	Retrodiction
Extension tier — depends on the φ-dark-matter sector				
$m_\phi = \Sigma m_\nu^\mathrm{act}(\mathrm{SOLAR}^2 K_v)$	φ-DM	$40.70\,\mathrm{eV}$	P6	**Prediction**
k_fs (Dodelson–Widrow)	φ-DM	$0.754\,h\,\mathrm{Mpc}^{-1}$	P6	**Prediction**
S_8^eff (two-sector)	φ-DM	0.758 (0.04σ KiDS)	P6	Retrodiction
$M = (5\varphi^8\rho_\mathrm{crit})^{1/4}$	UV compl.	$9.68\,\mathrm{meV}$	P10	Derived (UV)
Lensing (canonical EFT)	Strong gravity	GR+DM (uncond.)	P8	Retrodiction
Measured input (not derived)				
$A_s,\ \tau$	CMB amplitude	sampled ($k=2$)	P1,3	Measured
ρ_vac (abs. scale)	Vacuum	single measured input	P1,10	Measured

Table 5: Suite-wide master register: one row per observable across the ten-paper programme. "Postdiction": algebraic result fixed by the closed dictionary, then compared to already-public data (no claim of temporal priority). "Retrodiction": derived independently, then compared to an existing measurement. "Prediction": fixed now, tested against not-yet-released data. "Measured": an input, not a derivation. "Internal check": a same-paper consistency quantity, not itself tested against external data. No optimisation over the algebraic formula space was performed. The bare-field $\alpha_K^\mathrm{field} = 0.073$ (Paper 7) is the pre-integration kinetic contribution, distinct from the background-integrated $\alpha_K(0) = 0.4033$ used and cross-checked against `hi_class` in §4.1: both are Paper 7 quantities at different stages of the same calculation, not competing values for the same object. The detailed per-sector registers follow below and in the Predictive Register of Paper 1.

a gravitational-wave speed identical to light ($c_T = c$) to all orders in the dark sector, no running of the effective Planck mass, and no braiding. Any measured departure—an anomalous standard-siren distance, a time-varying G_eff, or a non-zero braiding signature in the ISW–galaxy cross-correlation—falsifies the canonical EFT presented in §4.

7 Honest accounting and outstanding challenges

A framework is only as credible as its honesty about what it does not yet explain. We therefore state plainly the inputs the model still requires, the exact parameter count, and the principal open problems—not as caveats buried in a footnote, but as the research agenda that the present work both motivates and constrains.

7.1 What is measured, not derived: the vacuum scale

SSEE does not predict the absolute energy density of the dark sector. The overall scale of the vacuum, $\rho_\Lambda \simeq 6 \times 10^{-10}\ \mathrm{J\,m^{-3}}$, enters as a single measured input, which we label *Postulate D*. What the algebra fixes is the *shape* of the dark-energy sector—its equation-of-state ratios w_0, w_a, the two-Ω structure, and the EFT function profile—not its amplitude. We make no claim to

Observable	SSEE value	Algebraic form	Falsifying probe
w_0	-0.83995	$-T_r/M_v$	DESI Y3 / Euclid BAO+SNe
w_a	-0.66997	$-P_{\rm sc}/I_g$	DESI Y3 / Euclid BAO+SNe
n_s	0.96556	$1-\varphi^{-7}$	Planck / CMB-S4
r	8.13×10^{-3}	φ^{-10}	LiteBIRD / CMB-S4
$\alpha_K(0)$	0.40330	$3\,\Omega_{\rm DE}\,\Omega_{\rm m}^{\rm dyn}$	DESI / Euclid growth
$\alpha_T, \alpha_M, \alpha_B$	0	exactly zero	GW standard sirens, ISW

Table 6: Core prediction register. Each value is fixed by φ and π alone, with no free parameter available to absorb a discrepancy. The tensor ratio $r=\varphi^{-10}$ sits an order of magnitude below the current bound $r<0.036$ and is the cleanest single falsification target: a detection above $\sim 10^{-2}$, or a robust upper bound below $\sim 8\times10^{-3}$, refutes the inflationary sector.

have solved the cosmological-constant problem; we have moved its free numbers from six tuned cosmological parameters down to a single measured normalisation plus the dimensionless invariants φ and π. This is a reduction in arbitrary inputs, not their elimination.

7.2 Parameter count: $k=2$ versus 6

We claim a *minimal-parameter* framework, never a zero-parameter one. The honest accounting is the following. ΛCDM carries six fitted parameters $(\Omega_b h^2, \Omega_c h^2, \theta_*, \tau, n_s, A_s)$. SSEE fixes four of them algebraically from φ and π: the baryon density ω_b (OP-1), the CDM density via the forward identity $\omega_c = \mathrm{KAL}_0\,\omega_b\,n_s$, the spectral index $n_s = 1-\varphi^{-7}$, and the expansion anchor $H_0 = 67.96\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ (whose value in these units coincides with $3(\varphi+\pi)^2$; derived, Paper 9). This leaves exactly two free CMB parameters—the primordial amplitude A_s and the optical depth τ—a $k=2$ versus $k=6$ comparison, which is the count used in the Paper 3 Bayesian model selection. The dark-energy ratios (w_0, w_a) and the EFT functions are predicted, not fitted. On the dark-sector extension we under-promise: the φ-DM mass coefficient (OP-9) and the non-minimal coupling ξ (OP-11) remain ansätze, tracked separately as open problems rather than counted among the fit parameters.

7.3 Open problems as bridges to future work

Three open problems are load-bearing and we name them explicitly. They define the path from the present minimal-parameter framework toward a fully derived theory.

- **The CMB matter density's algebraic origin (OP-1).** With the former MIRA matter bridge *dissolved* (the ω_m-direct reframe, §2.4; OP-8 retired), the CMB matter fraction $\Omega_{m,\rm CMB} = 0.30889$ now rests directly on two algebraic inputs: the baryon density $\omega_b = (\pi-\varphi)/[3(\varphi+\pi)^2] = 0.02242$ and the forward identity $\omega_c = \mathrm{KAL}_0\,\omega_b\,n_s = 0.11951$ (0.41σ from Planck). The CLASS Boltzmann test of §4 confirms this full density is *numerically necessary*—the acoustic peaks degrade by a factor of ~ 22 in RMS with the dynamical $\Omega_{m,\rm dyn} = 0.160$ alone. What remains open is a first-principles derivation of these algebraic forms from the field action; the burden has moved from "derive MIRA" to "ground ω_b and the ω_c identity." The values are not fitted, but their UV origin is open.
- **The φ-dark-matter mass (OP-9).** The mass $m_\varphi \simeq 40.70$ eV used in the extension sector follows from the algebraic relation $m_\varphi = \Sigma m_\nu^{\rm active}\,(\mathrm{SOLAR}^2\cdot\mathrm{KRYSTOS}_V)$, a dimensionally consistent product of a neutrino mass scale and a pure (φ,π) number. What remains open is the UV origin of the multiplier—a first-principles derivation from the field potential. The growth-sector predictions that descend from it—notably a free-streaming step in the matter power spectrum near $k_{\rm fs} \simeq 0.754\,h\,\mathrm{Mpc}^{-1}$, testable by DESI Y3 and Euclid—are deferred to

future work and deliberately excluded from the core register of Table 6 until the multiplier is derived.

- **The $\beta_c = -\mathrm{AURA}$ coupling (OP-7).** The conformal coupling agrees with $-\mathrm{AURA} = -3.998$ to 0.2%. We present this as a near-coincidence to be tested, not as a proven identity, pending a QFT-level derivation of the role assignments.

The growth-of-structure status of §5 mirrors this structure: the single-sector core inherits an S_8 tension at the single-sector ceiling (3.5σ KiDS-1000, $S_8 = 0.846$), and the two-sector extension that resolves it ($S_8 = 0.758$, 0.04σ of KiDS-1000) rests on the m_φ multiplier whose UV origin is the open item above. The non-linear treatment required to settle the matter definitively demands N-body simulations beyond this work's scope. We record the core-level tension as an open challenge and the extension as falsifiable future-data territory (DESI Y3, Euclid).

None of these gaps touch the algebraic core of Table 6: the dark-energy equation of state, the spectral index, the tensor ratio, and the vanishing EFT slip functions stand on the dimensionless invariants alone and are falsifiable today.

8 Conclusions

We have presented a minimal-parameter framework in which the shape of the dark sector is fixed by the two dimensionless constants φ and π. The late-time equation of state emerges as bare ratios of structural invariants, $w_0 = -0.840$ and $w_a = -0.670$, and matches the DESI DR2 posterior at 0.24σ (Pantheon+) without a single fitted dark-sector parameter. The effective-field-theory description is canonical, with vanishing tensor, running, and braiding functions and $\alpha_K(0) = 0.4033$; the inflationary sector predicts $n_s = 0.9656$ and $r = 8.1\times10^{-3}$; and a Boltzmann calculation confirms that the full ω_m-direct matter density (the two-Ω structure) is physically required rather than cosmetic.

The framework's strength is its frugality and its consequent falsifiability. Its core predictions—the dark-energy ratios, the spectral index, the tensor ratio, and the vanishing slip functions—rest on φ and π alone and will be tested decisively by DESI Y3, Euclid, LiteBIRD, and CMB-S4 within this decade. We have been equally explicit about its limits: the vacuum scale is a measured input, the S_8 growth amplitude remains in tension with weak lensing, and the algebraic origin of the CMB matter density (ω_b, ω_c) together with the φ-dark-matter mass coefficient are unresolved. These define a concrete research agenda rather than fatal flaws, and none of them touch the algebraic core.

Whether the deep agreement between a handful of golden-ratio combinations and the measured dark sector reflects an underlying structural principle or a remarkable set of coincidences is precisely the question that the prediction register of §6 is designed to settle. The next generation of cosmological surveys will provide the answer.

A The two lineage laws of the constant dictionary

The look-elsewhere test of §2.3 rests on the dictionary being a *closed, finite* set fixed in advance, rather than an open search space. This appendix states the generative rules that make it so. We present them as a descriptive grammar of the Genesis set [Almeida Vallejo, 2026]: they reproduce the dictionary exactly, but their *dynamical* origin is not claimed and remains the open mechanism of §7. Stating them explicitly is what turns "55 named invariants" from an assertion into a checkable construction.

Seeds and scaffold. Everything descends from two dimensionless seeds, $\varphi = (1+\sqrt{5})/2$ (UV) and π (IR). Their sum is the scaffold $\Omega = \varphi + \pi = 4.75963$, and their mean is the bifurcation point $\beta \equiv \mathrm{BIAL} = (\varphi + \pi)/2 = 2.37981$ at which the dictionary splits into two branches.

Law I — the copy law (φ-branch). The φ-branch is generated by a single root, the screening amplitude AURA $= \beta + \varphi = 3.99785$, which propagates *only* by integer multiples and the dyadic half. These are the dimensional ceilings:

$$\mathrm{MIRA} = \tfrac{1}{2}\mathrm{AURA}, \quad \mathrm{AURA}, \quad \mathrm{DUAL} = 2\,\mathrm{AURA}, \quad \mathrm{TRIAL} = 3\,\mathrm{AURA} = T_r, \quad \mathrm{CUARTAL} = 4\,\mathrm{AURA}. \tag{14}$$

The same law acts on the scaffold itself, giving the denominators $K_v = 2\Omega$ and $M_v = 3\Omega$. The economy noted in §2.3 is a direct consequence: since $T_r = 3\,\mathrm{AURA}$ and $M_v = 3\Omega$, the common factor of three cancels in

$$|w_0| = \frac{T_r}{M_v} = \frac{3\,\mathrm{AURA}}{3\,\Omega} = \frac{\mathrm{AURA}}{\Omega} = 0.83995, \tag{15}$$

so the equation-of-state numerator and denominator are not two independent draws but a single ratio of the two branch roots, AURA/Ω.

Law II — the no-self-sum law (π-branch). The π-branch is generated additively: each child is a sum of previously defined parents, subject to the constraint that *a node never adds to itself*. Thus KAL $= \beta + \pi$, P_{sc} (PYROS) $= \Omega + \varphi$, IGNIS $= \pi + P_{\mathrm{sc}}$, and so on, building up to the convergences. A corollary is that two distinct lineages may land on the *same value* without being the same entity: by construction IGNIS $= K_v = 9.51925$ (disruptive vs. structural lineage) and SOLAR $=$ MAR $= 7.90122$. The look-elsewhere count of §2.3 treats these honestly, deduplicating by value so the 55 named nodes collapse to 25 distinct numbers. The second equation-of-state ratio lives entirely on this branch,

$$|w_a| = \frac{P_{\mathrm{sc}}}{\mathrm{IGNIS}} = \frac{\mathrm{PYROS}}{\pi + \mathrm{PYROS}} = 0.66997, \tag{16}$$

where the denominator is read as the π-branch node IGNIS rather than the scaffold copy K_v—the two coincide in value by Law II.

Closure. The set is finite because both laws terminate at the four-dimensional integration invariant $M_v = \varphi + \pi + K_v = 14.27888$ (coinciding in value with 3Ω), the maximal dimensional invariant; no node exceeds it. This termination—not a choice made to fit data—is what bounds the dictionary to a fixed, enumerable set and licenses the conservative look-elsewhere denominator of §2.3. Any future extension would proceed only by the same copy law (the planned dimensional copies QUINTAL–DECAL), which, as noted in Table 2, add no accidental hit within the observational band of either w_0 or w_a.

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Data and Code Availability

The SSEE framework is fully transparent and reproducible. The complete code repository, including the CLASS and EFTCAMB configuration files, the COBAYA MCMC chains, the explicit record of Open Theoretical Problems (OP), and the cosmological validation ledgers, are publicly available at the GitHub repository: https://github.com/mikealmeida1721/SSEE.

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